AI ENGINEER INTERNSHIP CASE STUDY

Document Intelligence Platform

- End-to-end extraction, validation and reconciliation for financial documents.
- Live application: document-intelligence-platform-kace.onrender.com
- FastAPI | Gemini vision | Evidence-backed JSON

CASE-STUDY REQUIREMENTS

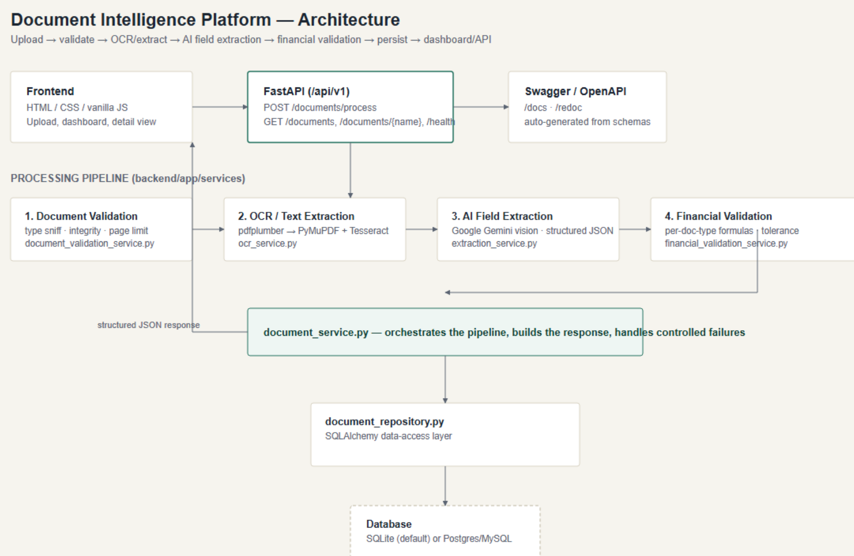
Objective and scope

- Four document types: invoice, balance sheet, profit and loss, cash flow statement.
- PDF/JPG/PNG uploads, maximum three pages and 15 MB, validated before OCR or AI.
- Meaningful fields and tables with source snippets and page evidence.
- Document-type financial formulas with PASS, FAIL and NOT_APPLICABLE.

System architecture

Browser dashboard -> FastAPI -> validation -> OCR -> Gemini vision -> financial validation -> repository -> database.

See architecture.png and architecture.pdf for the complete flow and service boundaries.



Validation, OCR and input safety

- File signatures, type, integrity, readability, page count and size are checked from bytes.
- Native PDF text is preferred; scanned pages use PyMuPDF rasterization and Tesseract fallback.
- Page-tagged text and normalized images preserve provenance for every field.

Evidence-backed AI extraction

- google-genai SDK, strict JSON envelope, line items, evidence and confidence.
- Missing or unreadable values stay null; no unsupported inference.
- Response variants, empty output, JSON fences, retry and model fallback are handled.

Financial validation engine

- Invoice: subtotal + tax - discount = total.
- Balance sheet: liabilities + equity = assets.
- Profit and loss: revenue - cost of sales = gross profit.
- Cash flow: operating + investing + financing + FX = net change.
- Rounding tolerance is explicit; missing operands produce NOT_APPLICABLE.

REVIEWABLE BY A HUMAN

Product experience and API

- Responsive dashboard, upload workflow, search, filters, statistics and refresh.
- Centered document workspace with Fields, Validation and Raw JSON tabs.
- POST /api/v1/documents/process; GET /api/v1/documents; GET /api/v1/documents/{name}; GET /api/v1/health; /docs.

EVIDENCE OF COMPLETION

Verification and deployment

- 25 automated tests pass without a network call or API key.
- Fresh live invoice verification returned file validation PASS and processing PASS.
- The live test returned 43 extracted field entries and 6 line items.
- Public Render deployment, health endpoint, Swagger and four-type sample fixtures are included.

READY TO SUBMIT

Deployment and submission links

- Backend API base URL: https://document-intelligence-platform-kace.onrender.com/api/v1
- Frontend URL: https://document-intelligence-platform-kace.onrender.com/
- Deployment platform: Render Docker web service.
- Public repository: https://github.com/SakthiPriyaR/document-intelligence-platform
- For a Google Drive PPT link, upload solution_presentation.pptx and set access to Anyone with the link - Viewer.

Limitations and production plan

- Render Free local SQLite storage is ephemeral; managed Postgres is required for durable history.
- Production adds authentication, tenant isolation, object storage, durable workers, observability and evaluations.

AI usage and submission deliverables

- AI coding assistance was reviewed and test-verified; Gemini is used only for configured, evidence-backed extraction.
- Repository includes source, live links, README, tests, samples, architecture source/PNG/PDF and this presentation.

SIX-MINUTE WALKTHROUGH

How to present the live demo

- Introduce the problem: turn unstructured financial files into traceable ledger records.
- Choose a document type, upload a supported document and explain validation before AI.
- Describe the path: OCR, Gemini extraction, arithmetic checks and persistence.
- Open the processed result to show fields, evidence, line items, validation and raw JSON.
- Close with deployment links, test evidence, limitations and production roadmap.